Original Article

Parallel-bar versus cross-bar configurations in the Nuss procedure: a systematic review and meta-analysis

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Background: Pectus excavatum (PE) represents the most prevalent congenital deformity of the chest wall, and minimally invasive repair of pectus excavatum (MIRPE) continues to evolve in pursuit of improved correction and reduced morbidity. Both cross-bar (CB) and parallel-bar (PB) configurations are widely adopted, yet comparative data remain limited. This systematic review and meta-analysis evaluated perioperative outcomes and postoperative efficacy of CB versus PB techniques.

Methods: Following PRISMA 2020 guidelines, PubMed, Embase, and the Cochrane Library were searched through October 24, 2025. Studies directly comparing CB and PB and reporting at least one predefined outcome were eligible. Pooled odds ratios (ORs) and mean differences (MDs) were estimated using a random-effects model based on the DerSimonian-Laird method. Risk of bias was evaluated using Risk of Bias In Non-randomized Studies of Interventions tool (ROBINS-I), and the certainty of evidence was graded according to Grading of Recommendations, Assessment, Development, and Evaluation (GRADE).

Results: Among the 1,021 patients analyzed, 449 underwent CB repair and 572 PB repair. Operative time [MD 4.79 min; 95% confidence interval (CI): −6.27 to 15.85] and length of hospital stay (MD 0.24 days; 95% CI: −0.39 to 0.88) did not differ significantly between techniques. Rates of pleural effusion [4.6% *vs.* 1.3%; odds ratio (OR) 3.59; 95% CI: 0.82–15.72] and pneumothorax (2.5% *vs.* 2.3%; OR 1.08; 95% CI: 0.39–2.97) were comparable between CB and PB. CB was associated with a small but statistically significant improvement in postoperative Haller Index (MD −0.10; 95% CI: −0.17 to −0.03). Sensitivity analyses revealed study-driven heterogeneity for operative time and hospital stay. Overall risk of bias ranged from moderate to serious, with low to very low certainty of evidence across outcomes.

Conclusions: CB and PB techniques demonstrate equivalent perioperative profiles and complication rates in MIRPE.

Keywords: Pectus excavatum (PE); minimally invasive repair of pectus excavatum (MIRPE); meta-analysis; postoperative outcomes

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Introduction

Pectus excavatum (PE), commonly referred to as "funnel chest", represents the most prevalent congenital deformity of the anterior chest wall, accounting for approximately 65–95% of such anomalies (1). Characterized by inward displacement of the sternum and adjacent costal cartilages, PE results in varying degrees of anterior thoracic depression (2). The condition typically presents in childhood or adolescence and is frequently associated with psychosocial distress, including poor body image, reduced social interaction, and decreased overall quality of life (3).

Although various surgical approaches have been developed, achieving complete and durable correction remains challenging, particularly in patients with complex, extensive, or asymmetric deformities. Historically, open techniques such as the Ravitch procedure were commonly employed, but the advent of the minimally invasive Nuss procedure significantly transformed the landscape of PE repair (4). Continuous refinements of the Nuss technique, including modifications in bar number, bar positioning, fixation methods, and adjunctive maneuvers, have aimed to improve correction quality, reduce complications, and enhance long-term outcomes (5).

Minimally invasive repair of pectus excavatum (MIRPE) may involve either single-bar or multi-bar internal support configurations. Among multi-bar approaches, the parallel-bar (PB) and cross-bar (CB) techniques are two commonly used strategies for improving substernal support and distributing corrective forces across the anterior chest wall. Among these, double-bar techniques have demonstrated improved outcomes, including shorter hospital stays and fewer complications compared with single-bar repairs (6). In the conventional PB configuration, bars are positioned in a horizontal fashion to provide broader substernal support and distribute corrective forces across the anterior chest wall (7). By contrast, the CB technique places the bars obliquely so that they intersect beneath the sternum, creating an X-shaped construct intended to improve multidirectional support, enhance correction of asymmetric or caudally depressed deformities, and increase overall stability (8). Notably, the configuration of these bars—whether parallel or crossed—may influence both mechanical stability and clinical results. CB techniques, particularly when combined with cryoanalgesia and stabilizing bridges, have shown comparable or even superior outcomes without increasing morbidity (9).

At the same time, the ideal number of bars and their overall arrangement remain debated among pectus surgeons. In his commentary on recent technical advances, Dr. Nuss acknowledged the value of crane-powered remodeling and modern bridge stabilization, but cautioned against the routine placement of additional bars, noting that in his experience only about 50% of patients require 2 bars and emphasizing the potential trade-off of greater dissection and hardware removal burden when extra bars are used unnecessarily (10). By contrast, Park and colleagues have advocated a morphology-tailored multiple-bar strategy aimed at comprehensive chest wall remodeling, using not only parallel and crossbar patterns but also XI configurations, and reported a progressive institutional shift away from single-bar repair (11). Thus, current practice is not defined by a universal gold standard, but rather by tailoring the number and arrangement of bars to deformity morphology, extent of chest wall involvement, and operative goals.

Nevertheless, the available evidence remains conflicting regarding whether one configuration is superior to the other, in part because of heterogeneity in study designs, technical implementation, and patient populations. To

Highlight box

Key findings

- In this meta-analysis of 4 retrospective studies (1,021 patients), cross-bar (CB) and parallel-bar (PB) configurations in minimally invasive repair of pectus excavatum (MIRPE) showed equivalent operative time, length of stay, pleural effusion, and pneumothorax. CB conferred a small but statistically significant reduction in postoperative Haller Index (MD −0.10; 95% CI: −0.17 to −0.03), under low to very low certainty of evidence.

What is known and what is new?

- Multi-bar constructs, both parallel and crossed, are widely used in MIRPE, and double-bar techniques outperform single-bar repair; however, direct comparative data on whether bar geometry itself influences perioperative outcomes have been limited and conflicting.
- This is the first systematic review and meta-analysis to pool head-to-head CB-versus-PB comparisons, demonstrating broadly equivalent perioperative and complication profiles, with only a modest—and clinically uncertain—anatomic advantage in Haller Index favoring CB.

What is the implication, and what should change now?

- Current evidence does not justify routinely preferring one configuration over the other based on perioperative outcomes alone. Technique choice should be individualized to deformity morphology, surgeon experience, and institutional pathways. Standardized prospective studies with longer follow-up and clearly defined endpoints (including recurrence, pain, and analgesic use) are needed to define comparative efficacy.

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address this knowledge gap, we performed a systematic review and meta-analysis comparing outcomes between PB and CB techniques in minimally invasive repair of PE. We present this article in accordance with the PRISMA reporting checklist (available at https://jtd.amegroups.com/article/view/10.21037/jtd-2026-1-0132/rc) (12,13).

Methods

Protocol and registration

The protocol was registered in PROSPERO (CRD420251183264) (14).

Eligibility criteria

Eligible studies were required to meet all of the following criteria for inclusion in the meta-analysis: (I) enrollment of patients diagnosed with PE; (II) design as a randomized trial or observational study; (III) direct comparison between the CB and standard PB techniques; and (IV) reporting of at least one outcome of interest. Studies were excluded if they (I) involved other surgical techniques, such as the sandwich technique, or (II) lacked a comparator group.

Search strategy and study selection

We conducted a systematic search of PubMed, Embase, and the Cochrane Library from database inception through October 24th, 2025. The full search strategies applied to each database are detailed in Table S1. In addition, we manually screened the reference lists of all included studies as well as relevant prior systematic reviews and meta-analyses to identify any additional eligible studies (15). Two authors (M.d.S.R. and E.C.B.) independently performed the literature search, imported all retrieved records into Rayyan (a web-based systematic review management platform), and carried out the study selection process. After removing duplicate records and excluding studies clearly unrelated to the research question based on title and abstract screening, we assessed the full texts of the remaining articles for eligibility. Any disagreements during the selection process were resolved through discussion and consensus.

Data extraction

Two authors (L.M.D. and G.C.S.) extracted the data from the included studies into a standardized format, including: general data (first author, year of publication, study design, and country), number of participants, patient characteristics [age, sex, body mass index (BMI), and Haller Index (HI)], follow-up, and outcomes reported.

Outcomes

The outcomes of interest were: (I) pleural effusion, (II) length of hospital stay (LOS), (III) operative time, and (IV) HI post procedure. Other outcomes were systematically collected and qualitatively summarized, even when quantitative pooling or graphical representation was not feasible.

Risk of bias and certainty assessment

We assessed the risk of bias using the Risk of Bias In Non-randomized Studies of Interventions tool (ROBINS-I, version 2) (16). Two reviewers (L.M.D. and G.C.S.) independently performed the assessments, and any disagreements were resolved through discussion and consensus. Owing to the limited number of included studies, publication bias could not be reliably evaluated, as funnel plot–based methods lack sufficient power to distinguish between random variation and true asymmetry when fewer than 10 studies are analyzed (17). We evaluated the certainty of the evidence using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach, classifying the overall certainty as high, moderate, low, or very low (18).

Statistical analysis

We applied a DerSimonian-Laird random-effects approach to pool effect estimates, calculating odds ratios (ORs) for binary outcomes and mean differences (MDs) for continuous outcomes, each reported with 95% confidence intervals (CIs) (19). Statistical significance was defined as a two-sided P value <0.05. Between-study heterogeneity was evaluated using the Cochran Q test and the I^2 statistic, with heterogeneity considered significant at P<0.10 or I^2>25%. When substantial heterogeneity was detected, we conducted leave-one-out sensitivity analyses to identify influential studies and assess their impact on the pooled results. All statistical analyses were performed using R statistical software (version 4.3.3; R Foundation for Statistical Computing, Vienna, Austria).

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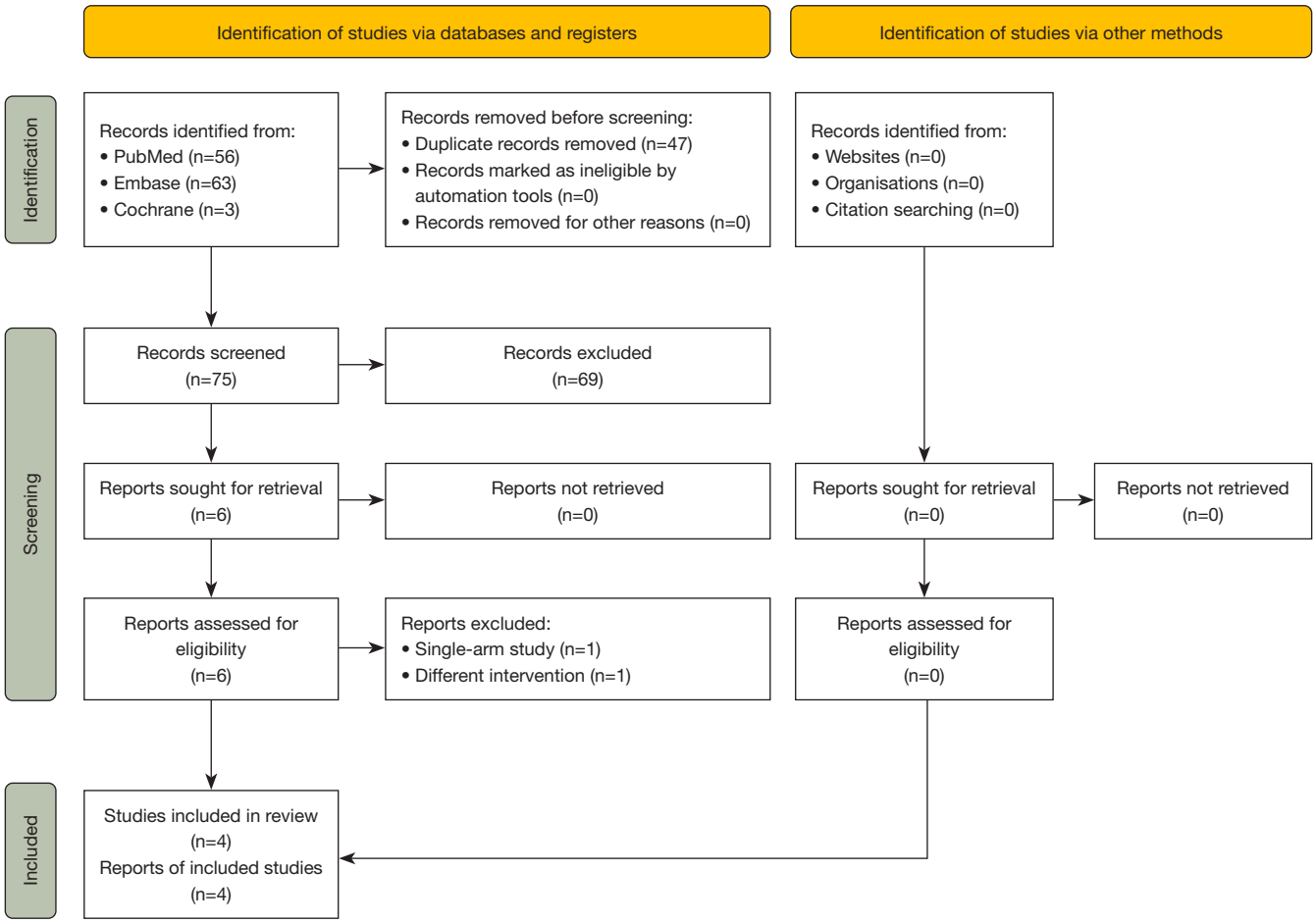


Figure 1 PRISMA 2020 flow diagram showing study identification, screening, and inclusion.

Results

Study selection

As illustrated in *Figure 1*, the initial literature search yielded 122 records. Following duplicate removal and title and abstract screening, 6 studies were retained for full-text assessment in accordance with the prespecified eligibility criteria. Of these, 4 retrospective observational studies, spanning from 2020 to 2023, were included (9,20-22). Two studies were excluded during the full-text screening stage: the study by DiFiore *et al.* lacked a comparator group (23), and the study by Park *et al.* evaluated the total crane lift technique (24).

Study and patient characteristics

A total of 1,021 patients were included across the four studies, with 449 (43.98%) undergoing the CB technique and 572 (56.02%) undergoing the PB technique. 80.2% were male, with a mean age of 20.5 years (20.8 years in the CB group and 20.3 years in the PB group). The mean BMI was 20.3 kg/m^2 in both groups, and the mean preoperative HI was 4.6 in the CB group and 4.5 in the PB group. Only Ersöz *et al.* reported the mean number of bars used in each group (9). The main demographic and clinical characteristics are summarized in *Table 1*.

Table 2 summarizes the technical characteristics of the compared procedures. Overall, both PB and CB configurations were variably implemented, supporting that the comparison was not between two fully uniform interventions. PB generally involved transverse placement of two or more bars, most commonly with a second bar positioned above the deepest point of the deformity, whereas CB used diagonal bar placement with intersection over or

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Table 1 Baseline demographic and clinical characteristics of patients undergoing minimally invasive repair of pectus excavatum using the cross-bar or parallel-bar technique across the included studies

First author, year	Design	Country	Period	No. of patients		Male, n (%)		Age (years), mean ± SD		BMI (kg/m^2), mean ± SD		Mean Haller Index		Follow-up time, months
				CB	PB	CB	PB	CB	PB	CB	PB	CB	PB	
Ersöz, 2024	R-Obs	Turkey, Austria, Israel and Russia, multicenter	2012–2022	205	213	176 (85.9)	169 (79.3)	23.9±8.4	22.6±8.1	21.9±1.8	21.4±1.4	4.3±2.0	4.48±2.1	NA
Hyun, 2023	R-Obs	South Korea, single-center	2016–2019	157	90	135 (86.0)	68 (75.5)	18.0±6.1	15.7±5.0	18.7±2.6	18.5±2.7	5.0±2.1	4.5±1.1	Median: 40.8
Moon, 2020	R-Obs	South Korea, single-center	2015–2018	36	44	31 (86.1)	36 (81.8)	17.0±4.5	16.9±3.9	18.0±2.1	18.9±2.6	4.9±1.5	4.5±1.0	NA
Sayan, 2022	R-Obs	Turkey, single-center	2005–2018	51	225	45 (88.2)	159 (77.6)	20.1	20.7	NA	NA	NA	NA	24–36

BMI, body mass index; CB, cross-bar; NA, not available; PB, parallel-bar; R-Obs, retrospective observational; SD, standard deviation.

Table 2 Technical characteristics of parallel-bar and cross-bar procedures across the included studies

First author, year	Intervention configuration		Technical variations		Stabilization methods	Sternal elevation	Crane/vacuum device	Hammock stitches	Bar shaping/twisting	Thoracoscopic approach	Degree of mediastinal dissection	Surgeon experience
	PB	CB	PB	CB								
Ersöz, 2024	Two or more bars placed without any intersection	Two or more bars placed diagonally so that they intersect at any point	Transverse bar placement at the point of maximal depression, with additional bars positioned superiorly as needed; individualized bar shaping and rotation	Diagonal bar insertion across intercostal spaces (e.g., upper-right to lower-left and vice versa), creating an "X" configuration; individualized bar shaping, rotation, and twisting to optimize lateral chest wall support	Bar stabilizers used in both groups; additional stabilizers applied in cross-bar configurations as needed. No fixation to surrounding tissues; stability achieved through bar contouring and sternal compression	Yes	Crane or vacuum device	NA	Yes	Right-sided thoracoscopy with CO_2 insufflation; direct visualization	Right-sided thoracoscopy with CO_2 insufflation; direct visualization	NA
Hyun, 2023	First bar placed transversely at the deepest point; second bar set parallel 1–2 intercostal levels above	Two bars placed diagonally to cross at the center/target; 33% received an additional upper bar in XI fashion	Horizontal double-bar support for broader deformities, especially broad chest wall depression/Grand Canyon type	Diagonal crossing to concentrate lifting force on a focal target and extend coverage to lower/lateral chest wall; additional upper bar in XI fashion when upper chest lift was insufficient; sandwich-derived adjuncts for mixed deformities were described but excluded from this comparative cohort	Bridge plate fixation with connecting metal plates and tight screw fixation in both groups	Yes	Crane technique (table-mounted crane elevator)	NA	Bars bent according to chest wall contour using the terrain contour-matching principle	Right-sided thoracoscopic guidance using the pectoscope for direct visualization	Described: mediastinum dissected in a right-to-left direction; second CB bar shared the same mediastinal tunnel beneath the first bar	NA
Moon, 2020	First bar inserted transversely at the deepest point; second bar placed transversely 1–2 intercostal spaces above	Two bars inserted diagonally crossing at the target area (upper and lower intercostal spaces), forming an "X" configuration	Double-bar transverse configuration to cover multiple rib levels and distribute force along the chest wall	Crossing mechanism creates a lever effect to lift both the deepest point and broader chest wall, particularly effective for lower/caudal deformities	Bridge plate fixation used to secure bar ends in both groups	Yes	Crane technique (sternal elevation with wire sutures)	NA	Bars configured according to individual chest wall morphology	Thoracoscopic guidance implied but not explicitly detailed	NA	NA
Sayan, 2022	Parallel bars placed transversely at levels corresponding to the deformity; number of bars based on severity	Cross-bar configuration with bars inserted obliquely across intercostal spaces, forming an intersecting construct	Standard parallel Nuss configuration adapted to deformity extent; multiple bars used when needed	Oblique bar placement designed to provide broader support across the chest wall; intended to improve stability in more complex deformities	Stabilizers used; details on fixation technique not fully specified	NA	NA	NA	NA	Thoracoscopic guidance implied but not detailed	NA	NA

CB, cross-bar; NA, not available; PB, parallel-bar.

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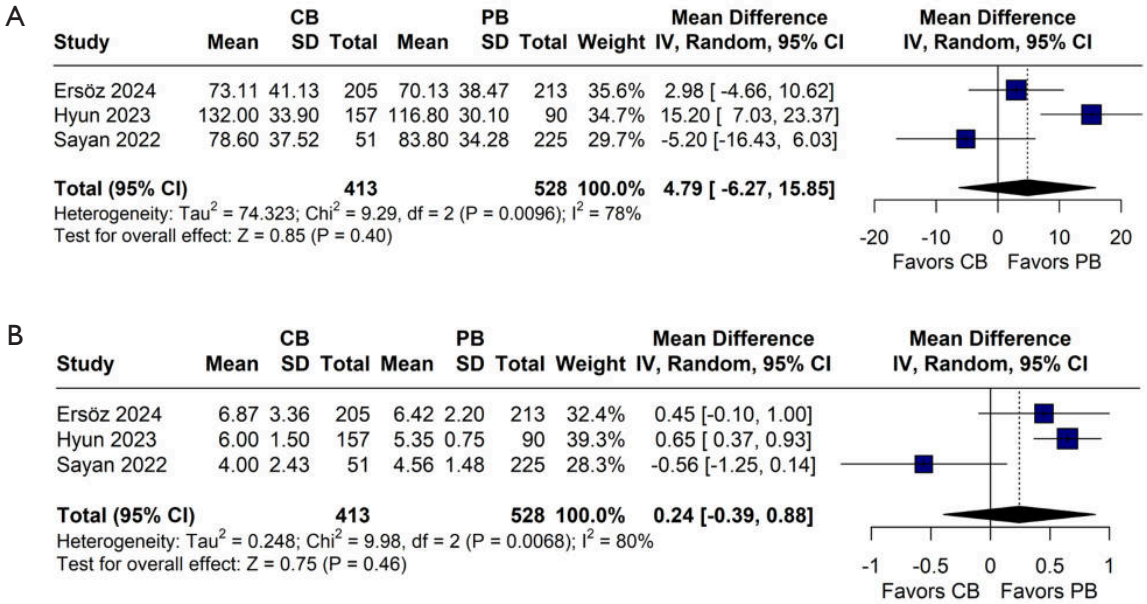


Figure 2 Forest plots for time-related outcomes comparing CB and PB configurations in MIRPE. (A) Operative time (minutes); (B) length of hospital stay (days). CB, cross-bar; CI, confidence interval; IV, inverse variance; MIRPE, minimally invasive repair of pectus excavatum; PB, parallel-bar; SD, standard deviation.

near the target area. Bridge plate or stabilizer-based fixation was the most commonly reported stabilization strategy, although the level of detail varied. Sternal elevation was reported in three studies, usually with crane-assisted lifting, while hammock stitches were not reported in any included study. Thoracoscopic guidance was consistently mentioned when technical details were provided, most explicitly as right-sided thoracoscopy with direct visualization in Ersöz and Hyun (9,20). In contrast, the degree of mediastinal dissection and surgeon experience were seldom described.

Analgesic strategies varied substantially, with the routine use of intravenous patient-controlled analgesia (PCA) in some cohorts (9,21,22), and continuous intercostal nerve catheter techniques in others (20), alongside variability in duration and transition to oral medication. Notably, none of the included studies reported the use of cryoanalgesia, and no standardized enhanced recovery after surgery (ERAS) pathways were described. Additionally, formal comparative assessment of pain outcomes between techniques was lacking. These elements are detailed in Table S2.

Pooled analysis

Time-related outcomes

Three studies reported operative time and LOS (9,20,22). There were no statistically significant differences between groups for operative time (MD 4.79 minutes; 95% CI: −6.27 to 15.85; P=0.40; I^2=78%; *Figure 2A*), and for LOS (MD 0.24 days; 95% CI: −0.39 to 0.88; P=0.46; I^2=80%; *Figure 2B*). However, discharge criteria and institutional pathways were poorly reported. None of the included studies explicitly defined discharge criteria in terms of pain control, mobility, or imaging requirements, and no standardized postoperative pathways were described. Thus, although LOS was consistently reported as an outcome, it was not contextualized within protocol-driven discharge criteria. These factors, summarized in Table S3 may represent important sources of variability and could explain the heterogeneity observed for perioperative outcomes.

Postoperative complications

Only pleural effusion and pneumothorax were reported by two or more studies. There were no statistically significant differences between groups for pleural effusion (OR 3.59; 95% CI: 0.82 to 15.72; P=0.09; I^2=58%; *Figure 3A*), and for pneumothorax (OR 1.08; 95% CI: 0.39 to 2.97; P=0.89; I^2=0%; *Figure 3B*). Ersöz *et al.* defined pneumothorax as cases requiring postoperative drainage (9), whereas Hyun *et al.* did not further specify the criterion used (20).

Ersöz *et al.* reported one pericardial effusion in the CB

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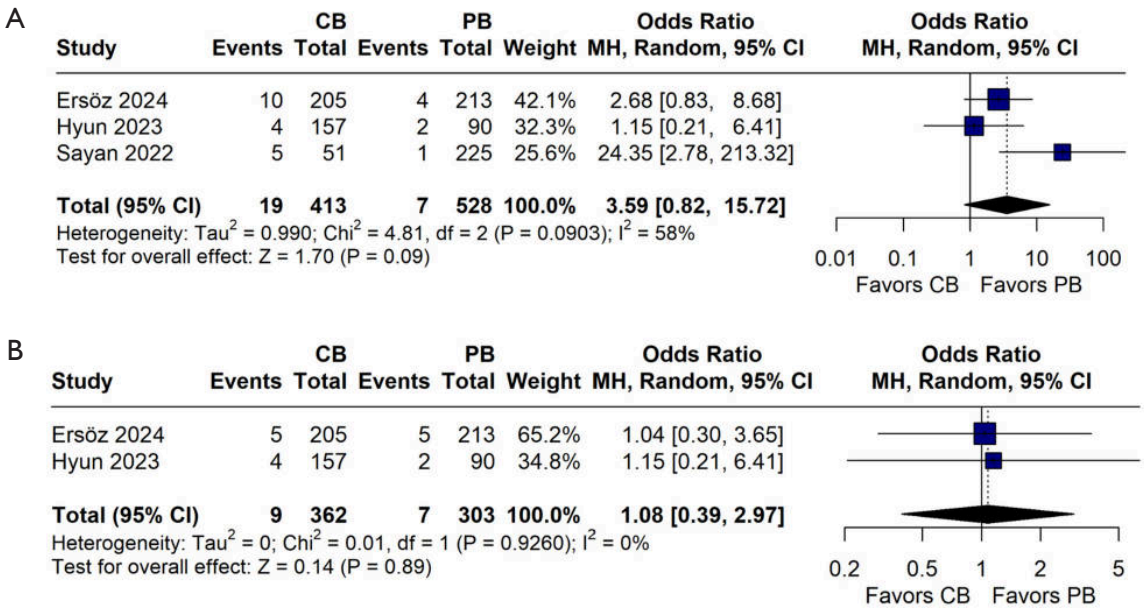


Figure 3 Forest plots for postoperative complications comparing CB and PB configurations. (A) Pleural effusion; (B) pneumothorax. CB, cross-bar; CI, confidence interval; MH, Mantel-Haenszel; PB, parallel-bar.

group, fewer bar shifts with CB (4 *vs.* 2), similar rates of persistent chest pain (7 *vs.* 8), and markedly lower recurrence after CB repair (1 *vs.* 7) (9). In addition, in the study by Hyun *et al.*, hemothorax occurred at the same rate in both groups (1 case each), while brachial plexopathy was observed only in the CB group (2 cases) and wound infection was slightly higher with CB (6 *vs.* 4 cases) (21). The study by Moon *et al.* did not report any complication (21).

Reintervention and reoperation outcomes were inconsistently reported. Ersöz *et al.* reported reoperations linked to bar shifts only, with 4 events in the PB group and 2 in the CB group. Hyun *et al.* reported no reoperation in either group during follow-up. Moon *et al.* did not report reintervention or reoperation outcomes. Sayan *et al.* reported hospital readmission due to serous pleural effusion more frequently in the CB group than in the PB group (5 *vs.* 1), but these events were described as readmissions managed with drainage rather than as a standardized reoperation endpoint.

Efficacy-related outcome

Postoperative HI was reported by two studies (20,21). CB significantly reduced HI compared to PB (MD −0.10 points; 95% CI: −0.17 to −0.03; P<0.01; I^2=0%; *Figure 4*). As detailed in *Table 3*, baseline HI was slightly higher in the CB group in both studies, while postoperative HI values were similar between groups. Consistent reductions from baseline to postoperative HI were observed in both CB and PB groups.

Patient-reported and cosmetic outcomes

Only Ersöz *et al.* reported a patient-reported satisfaction outcome (9), assessed with a modified single-step questionnaire adapted from Krasopoulos *et al.* and focused on satisfaction after bar removal (25). PB showed significantly higher values compared to CB (9.40±0.95 *vs.* 9.13±1.11, P=0.01). The remaining studies did not report any patient-reported outcomes.

Pain-related outcomes

Ersöz *et al.* reported persistent chest pain at 6 months, with similar rates between PB and CB (3.3% *vs.* 3.9%), and noted routine postoperative use of intravenous (IV) patient-controlled analgesia (PCA) (9). Hyun *et al.* reported the use of local infiltrative intercostal nerve anesthesia catheters, but pain differences between groups were not assessed (20). Moon *et al.* only stated that IV PCA was routinely initiated, without comparative pain data (21). Sayan *et al.* described postoperative analgesia with self-controlled intravenous morphine for 3 days followed by oral analgesics, but also did not report comparative pain outcomes (22).

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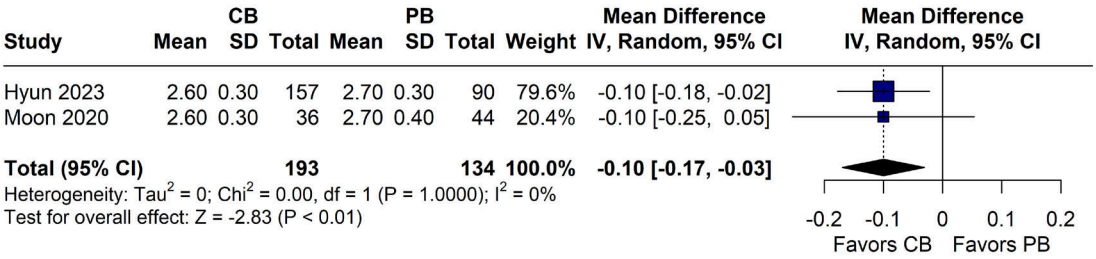


Figure 4 Forest plot for efficacy-related outcome showing postoperative Haller Index comparison between CB and PB techniques. CB, cross-bar; CI, confidence interval; IV, inverse variance; PB, parallel-bar; SD, standard deviation.

Table 3 Preoperative and postoperative Haller index according to bar configuration

First author, year	Baseline HI, mean ± SD		Post-operative HI, mean ± SD		ΔHI	
	CB	PB	CB	PB	CB	PB
Ersöz, 2024	4.3±1.99	4.48±2.08	NA	NA	NA	NA
Hyun, 2023	5.0±2.1	4.5±1.1	2.7±0.3	2.6±0.3	−2.3	−1.9
Moon, 2020	4.9±1.5	4.5±1.0	2.7±0.4	2.6±0.3	−2.3	−1.9
Sayan, 2022	NA	NA	NA	NA	NA	NA

CB, cross-bar; HI, Haller Index; NA, not available; PB, parallel-bar; SD, standard deviation.

Long-term outcomes

Ersöz *et al.* reported recurrence more frequently in the PB group than in the CB group (7 *vs.* 1; 3.3% *vs.* 0.5%), although the definition and timing of recurrence assessment were not standardized. Bars were left in place for at least 3 years before removal, but no post-removal outcome assessment was reported (9). In Hyun *et al.*, bar removal was scheduled according to age: after 2 years in patients aged ≤12 years, after 2.5 years in those aged 12–18 years, and after 3 years in patients older than 18 years. Bar removal was completed in 237 patients (96.1%) after a median interval of 35.7 months, with no recurrence observed after removal over a mean follow-up of 40.8 months (20). Sayan *et al.* also used age-based bar removal, scheduled at 2 years in patients younger than 17 years and at 3 years in those older than 17 years, but no comparative post-removal outcomes were provided (22). Moon *et al.* reported neither bar removal nor recurrence (21). Opioid analgesic consumption was not reported by any study.

Sensitivity analysis

Leave-one-out analysis is detailed in Figures S1-S3. For LOS, the exclusion of the study by Sayan et al. eliminated heterogeneity while shifting the results from nonsignificant to significant favoring PB (22). For pleural effusion, the exclusion of the same study also eliminated heterogeneity; however, it did not change the significance of the results. For operative time, the exclusion of any individual study significantly reduced heterogeneity.

Risk of bias and certainty assessment

As detailed in *Figure 5*, across the four included non-randomized studies comparing CB and PB techniques for MIRPE, the risk of bias ranged from moderate to serious. Most studies presented clear intervention definitions and objective radiologic endpoints, supporting low risk in the classification and outcome measurement domains. However, all were retrospective and lacked randomization; confounding by deformity severity, surgeon experience, and institutional protocol differences was the predominant limitation. Only Moon *et al.* performed limited multivariable adjustment (21), whereas the remaining studies relied solely on descriptive comparisons. Missing-data handling was generally adequate, but selective emphasis on significant results and absence of prespecified protocols yielded moderate reporting bias. Consequently, the overall risk of

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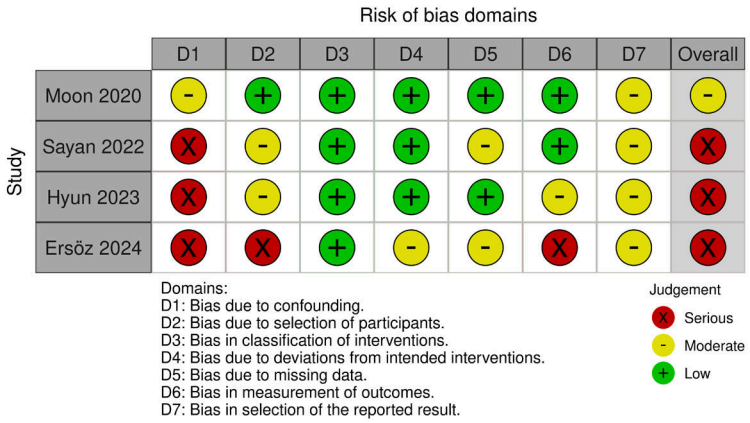


Figure 5 Risk of bias summary using the ROBINS-I tool for the included non-randomized studies comparing CB and PB configurations. CB, cross-bar; PB, parallel-bar.

bias judgment for the observational evidence was moderate-to-serious, indicating that residual confounding and non-standardized management could meaningfully influence effect estimates.

The GRADE certainty of evidence was low to very low across all outcomes due to study design limitations, imprecision, and inconsistency, as detailed in Table S4. All included studies were retrospective and non-randomized, with moderate to serious confounding risk. Additional downgrading was applied for inconsistency (substantial heterogeneity, I^2=58–80%) and imprecision (wide CIs crossing the null, small sample sizes, and low event counts).

Discussion

In this systematic review and meta-analysis of 4 studies (2020–2024; n=1,021) comparing CB versus PB configurations in MIRPE, we found: no differences in operative time, or LOS, comparable complications for pleural effusion and pneumothorax; and a small but statistically significant improvement in postoperative HI favoring CB. Sensitivity analyses showed study-driven heterogeneity for LOS and operative time; removing Sayan *et al.* reduced LOS heterogeneity and shifted the effect toward PB. These findings should be interpreted cautiously, given the small number of included studies, the observational nature of the evidence base, and the low to very low certainty of evidence across outcomes.

Beyond the direct comparison between PB and CB, our findings should be interpreted within the broader ongoing debate on the optimal MIRPE strategy, including both bar number and configuration. Nuss has cautioned against routine use of additional bars, noting that not all patients require multiple bars and that extra hardware may increase dissection and removal burden (10). By contrast, Park and colleagues have advocated a morphology-tailored multi-bar strategy, including parallel, CB, and XI configurations, to achieve more comprehensive chest wall remodeling (11). This contrast highlights the lack of robust comparative evidence in the field and suggests that current practice remains driven largely by surgeon experience, institutional preference, and deformity-specific considerations rather than by a universally accepted standard.

The lack of differences in operative time and LOS is consistent with comparative series in which institutional pathways, including analgesic strategy, use or absence of cryoanalgesia, discharge criteria, bar/stabilizer strategy, and sternal elevation, may have a greater influence on these outcomes than bar geometry itself (26,27). The shift toward PB in LOS after excluding a single study implies that center-level practices and technical preferences may dominate these process measures. Practically, ERAS-style standardization, multimodal analgesia, and unified discharge criteria probably modulate LOS more than CB versus PB per se (28-30). At the same time, variability in patient populations, operative strategies, and institutional protocols across studies further limits direct comparability of these pooled process outcomes.

Pooled pleural effusion and pneumothorax were comparable between CB and PB. The near-significant trend toward more effusions with CB mirrors cohort signals of increased serous effusions/readmissions in crossed

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constructs, possibly related to bar trajectories, contact points, and drainage pathways (31-33). Although this result did not reach conventional statistical significance, the elevated point estimate suggests that a clinically relevant increase in pleural effusion risk with CB cannot be excluded. Pneumothorax, low-frequency and multifactorial, remained stable across techniques. Rare events (e.g., brachial plexopathy with CB) appear operator/technique dependent (traction/positioning, bar count, bar rotation), supporting meticulous checks as mitigation (34,35).

The HI advantage with CB was small in absolute terms but consistent with the biomechanical rationale that bar convergence can redistribute forces, improve sternal rotation control, and enhance lower/lateral chest remodeling (11,23,36,37). Nevertheless, a –0.10 HI difference, though statistically significant, has context-dependent clinical relevance, varying with defect morphology (asymmetry, caudal rotation) and aesthetic/functional goals. Importantly, the magnitude of this difference is small and may not translate into a clinically meaningful improvement in patient outcomes or decision-making. Broadly, current evidence suggests perioperative equivalence of CB and PB, with a modest anatomic edge for CB. Since postoperative HI was reported in only two studies, any morphology-based advantage should be interpreted cautiously. Moon *et al.* suggested greater correction in the lower portion of the deformity (21), whereas Hyun *et al.* described CB as particularly suited for rigid, focal, and caudally located defects (20). Accordingly, this finding should be viewed as an anatomic signal rather than definitive evidence of overall clinical superiority.

Given similar operative, LOS, and complication profiles, along with only a modest HI benefit favoring CB, technique selection should be individualized and may reasonably remain guided by surgeon preference, intraoperative judgment, and patient-specific anatomy in the absence of consistent outcome superiority. Clinically, these findings do not support a universal preference for one bar configuration over the other based on perioperative outcomes alone. Rather, the choice of configuration should be tailored to deformity characteristics and local technical expertise, while recognizing that the small HI difference observed with CB may not be sufficient, by itself, to justify a change in practice. This rationale is supported by technical descriptions of the CB approach, which has been proposed particularly for complex or asymmetric deformities requiring broader and more multidirectional force distribution. Park described the technique for eccentric long canal ("Grand Canyon type") deformities, emphasizing its ability to elevate multiple targets and remodel the entire anterior chest wall rather than only the point of maximal depression (8). CB may be better suited to selected complex morphologies, particularly focal or caudally located deformities, whereas PB within robust ERAS frameworks may optimize LOS.

In addition, emerging technical descriptions suggest that CB configurations may be particularly useful in specific anatomical scenarios. These include cases in which the intercostal spaces do not align with the point of maximal sternal depression, potentially limiting effective elevation with a standard transverse bar. Furthermore, in patients with very inferior deformities—especially when the point of maximal depression is located near the xiphoid or below the bony sternum—the position of the diaphragm may preclude adequate placement of a conventional PB. In such cases, oblique CB configurations allow more precise targeting of the deepest point of deformity and may improve correction (11,23,36,37). Regardless of configuration, preoperative planning (bar/stabilizer number) and intraoperative safety checks (position, traction, axillary compression) are advisable to minimize neurologic events and seromas (38-40). However, because the available literature does not adequately address long-term recurrence, functional improvement, or patient-reported outcomes such as satisfaction, the overall comparative success of CB versus PB remains incompletely defined.

From a practical standpoint, technique selection should be guided by deformity characteristics and operative context rather than by a one-size-fits-all preference. CB configurations may be considered in asymmetric, caudally displaced, or more focal deformities requiring broader and multidirectional force distribution, whereas PB techniques may be favored in more symmetric defects and in settings with standardized perioperative pathways. Regardless of configuration, meticulous intraoperative planning, including bar positioning, fixation strategy, and careful attention to patient positioning and traction precautions, remains essential to optimize correction and to minimize rare neurologic complications (41-44).

This study has limitations. First, the evidence base is small, with only four published studies eligible for inclusion, which restricts the breadth of available data and prevents reliable assessment of publication bias and small-study effects; therefore, selective publication cannot be excluded and the comparative findings should be interpreted cautiously, although the total number of patients analyzed

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remains meaningful. Second, all included studies had observational designs, with an inherent risk of confounding related to factors such as deformity severity/asymmetry, surgeon experience, and protocol drift. To mitigate this, we used random-effects modeling and leave-one-out analyses to probe robustness. Third, there was substantial heterogeneity for LOS and operative time. Sensitivity analyses suggested that this heterogeneity was study-driven and did not overturn the overall finding of equivalence. Nevertheless, the residual heterogeneity remains an important limitation and reduces confidence in direct pooled comparisons for these outcomes. Fourth, several outcomes were imprecise because of low event rates and wide CIs for complications. Accordingly, these findings were interpreted conservatively, and sparse outcomes were synthesized qualitatively when appropriate. Fifth, comparability across studies was limited, as only one study reported multivariable-adjusted estimates. This was addressed through formal ROBINS-I appraisal and GRADE downgrading. Finally, publication bias could not be formally assessed because fewer than 10 studies were available, and standardized reporting of outcomes after bar removal and long-term recurrence remained limited. Several clinically relevant outcomes—such as reintervention/reoperation, recurrence, postoperative pain/analgesic burden, patient-reported outcomes, cosmetic satisfaction, and post–bar-removal outcomes—were also inconsistently reported, often lacked standardized definitions, and were assessed over heterogeneous follow-up periods, which precluded quantitative synthesis. The included studies also did not consistently report functional outcomes or patient-centered endpoints, further limiting the clinical interpretability of comparative efficacy. Future studies should adopt standardized outcome definitions and provide longer follow-up.

Conclusions

Among MIRPE patients, CB and PB demonstrate similar operative time, LOS, and common complications, while CB was associated with a small postoperative HI difference favoring this configuration. Given the low certainty of evidence, the limited number of included studies, and inter-center variability, current evidence suggests broad perioperative equivalence between CB and PB, and the observed HI difference should be interpreted cautiously. Clinically, these findings do not support routine preference for one configuration over the other based on perioperative outcomes alone. Taken together, these findings suggest that the choice between configurations is likely to depend more on surgeon preference, technical judgment, and patient-specific anatomical considerations than on clear and consistent differences in clinical outcomes.

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Footnote

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